Task 1 — Data Collection from the Video

1. Measurement Setup

The source video shows a general public crowd moving from **right to left** (mostly) across the screen. Occasional individuals walk in the opposite direction (left to right). Travelers carry little to no baggage, and their average walking speed is estimated at **1.5 m/s**. To extract reliable flow parameters, I performed a **frame‑by‑frame analysis** to count the arrival rate with a virtual measurement line on the left of the screen.

Virtual Gate Definition

A vertical "gate" line at the **left edge of the screen**. Any traveler who crosses the left screen edge from right to left is counted as **positive flow**; any crossing from left to right is counted as **counter‑flow**.

2. Observations

I counted the number of travellers crossing the gate in each direction.

Duration (seconds)	Right‑to‑left crossings	Left‑to‑right crossings	Total crossings
0:24	31	11	20

The flow appears steady with no significant congestion.

3. Derived Parameters

3.1 Arrival Rate (λ)

The main flow (right‑to‑left) crossing the left edge is equivalent to travellers **entering** the checkpoint area. Using the average count per second:
λ_observed = 31.0 travellers / 24 seconds = 1.3 travellers/second
λ = 1.3 travellers/s (78 travellers/min)

3.2 Direction Proportion

- **Main flow** (right‑to‑left) = 95% of arrivals
- **Counter‑flow** (left‑to‑right) = 5% of arrivals

In the simulation, we will exclude the counter flow.

3.3 Walking Speed

The average speed is about **1.5 m/s**. We validated this by tracking a few individual travellers across an estimated distance in the video with the stride of the travellers. The simulation will use the base speed with variations by traveler type and triangular random to be natural:
v_base = 1.5 m/s

3.4 Traveller Type Distribution

All travellers appear **lightly laden** with no noticeable variation in walking speed or luggage. However, the simulation will use **3 traveller types** (“standard”, “business”, “family”). It is configurable to 100% “standard” if the scenario in the video needs to be simulated. To introduce realistic variation in processing times (security & immigration), we will also apply a **random triangular distribution** with configurable min, mode, and max.

4. Simulation Parameters Summary

Parameter	**Value**	**Notes**
Arrival rate (main flow)	1.3 persons/s	Poisson process
Average walking speed	1.5 m/s	Constant for all travellers
Traveller type	Standard only	No business / family distinction
Processing time variation	Triangular(min, mode, max)	e.g., security: 30/60/90 s

5. Assumptions

1. **Peak‑state flow** – The observed 24‑second window is representative of the rush‑hour peaks.
2. **Gate placement** – The left edge represents the exit of the checkpoint; therefore arrivals at the checkpoint entrance equal the observed flow.
3. **No queuing upstream** – The video shows free‑flow walking; we assume travellers do not queue before the checkpoint entrance.
4. **Counter‑flow behaviour** – For simplicity, we will exclude counter-flow behaviour in the simulation.
5. **Walking speed constant** – No acceleration/deceleration; instant step change when direction is assigned.